統計模擬報告 : Markov’s inequality

第六組:張肇麟、張智賢、黃柏翰、賴元中

大綱

01

介紹不等式

02

違反條件之分配

03

不同分布嚴謹程度

04

總結

01

介紹不等式

Markov's inequality

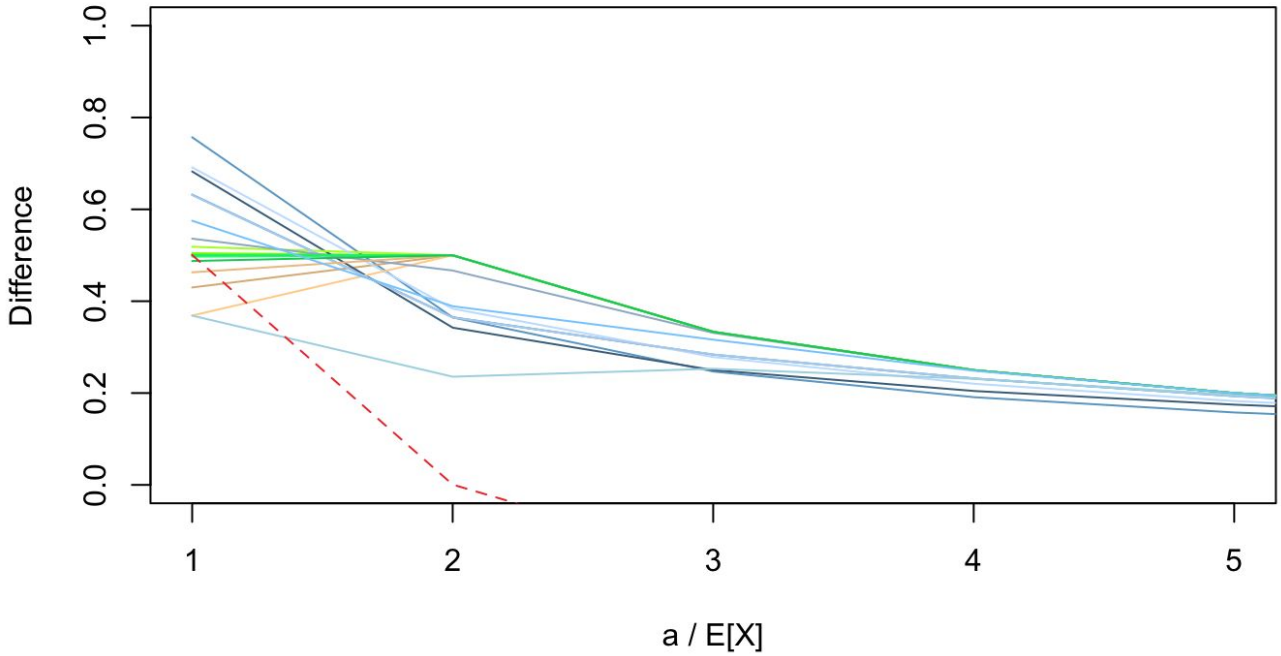
$$P(X \geq a) \leq \frac{E(X)}{a}$$

- **P(X ≥ a)**: is the probability that X is greater than or equal to 'a'.
- **E(X)**: is the expected value of the random variable X.
- **a**: is a positive real number.
- **X**: is a non-negative random variable.

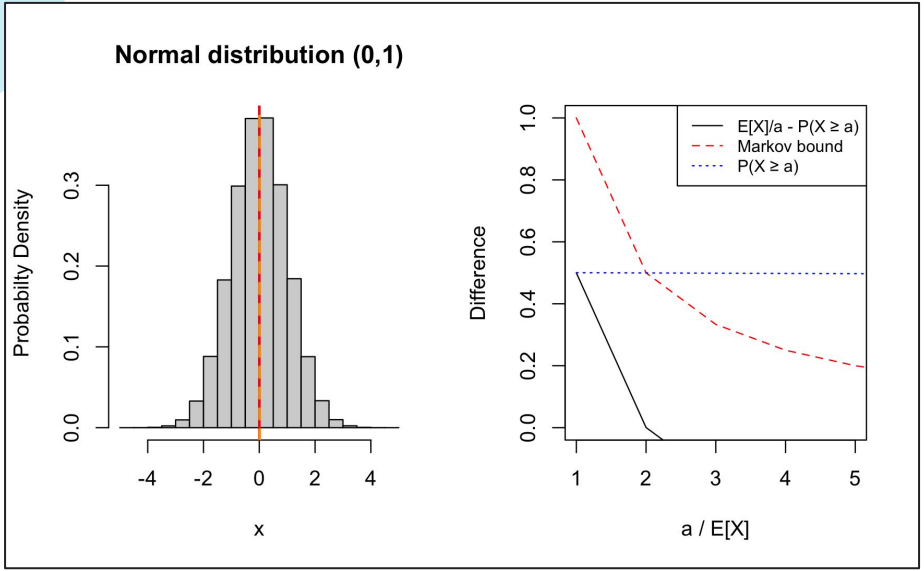
02

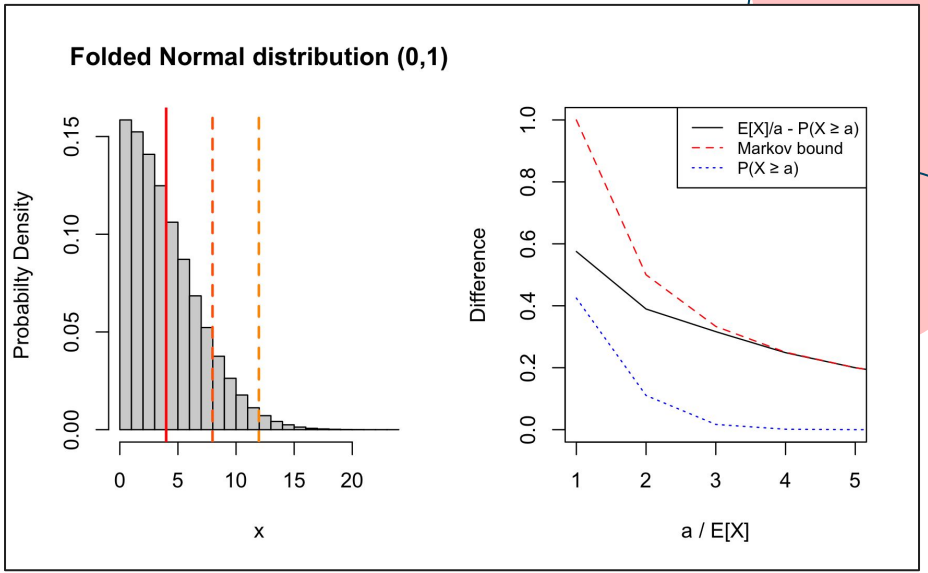
違反條件之分配

$$P(X \geq a) \leq \frac{E(X)}{a}$$



Normal Distribution

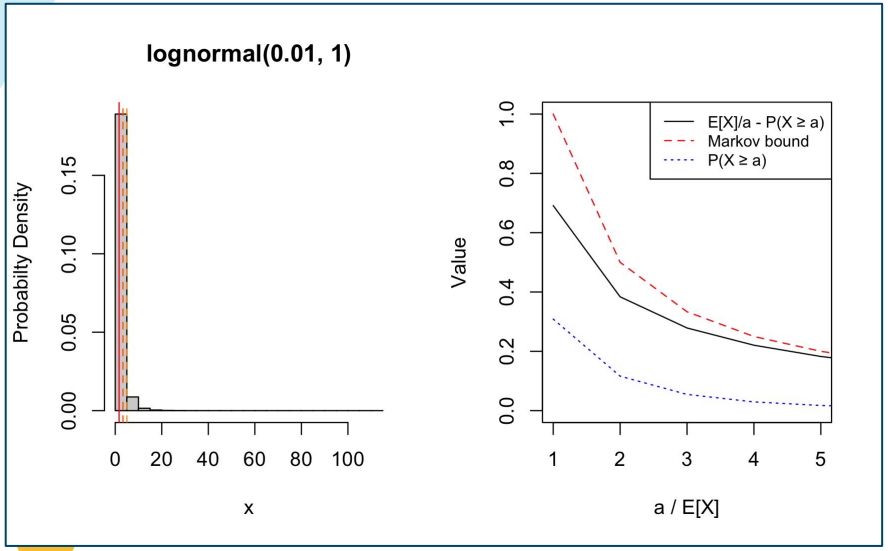


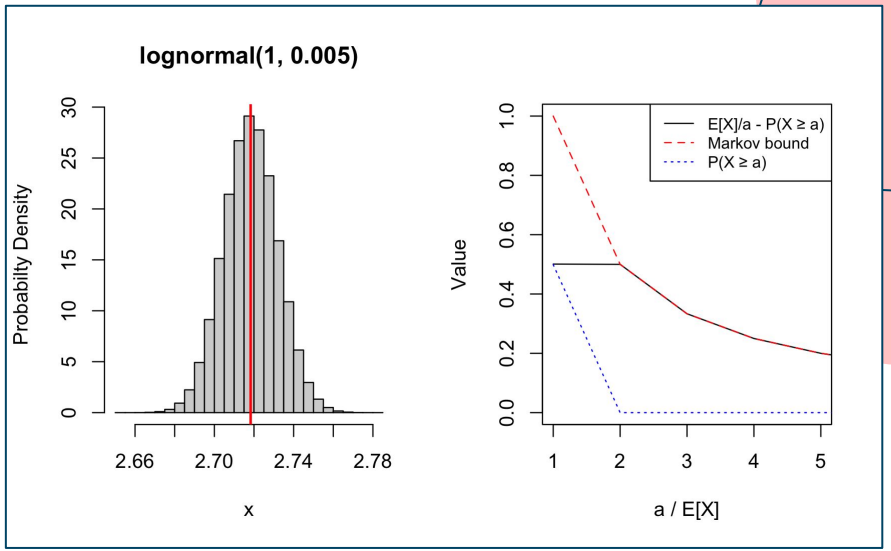


03

不同分布嚴謹程度

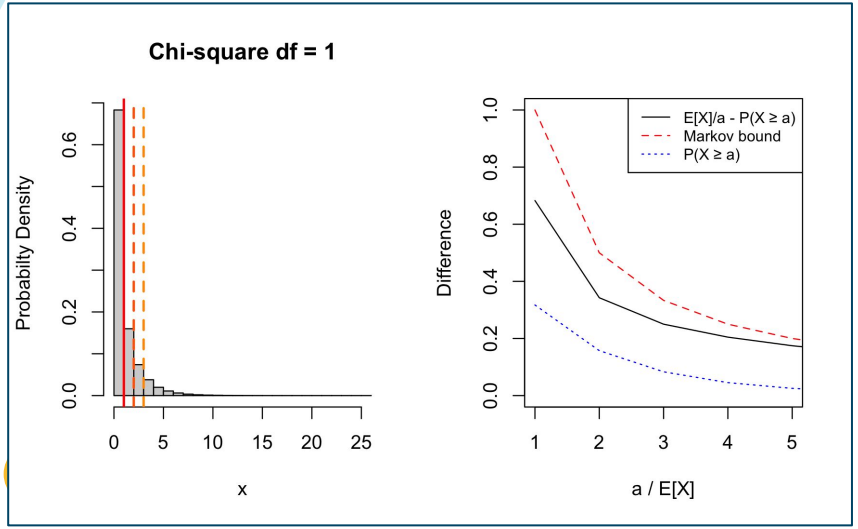
Lognormal Distribution



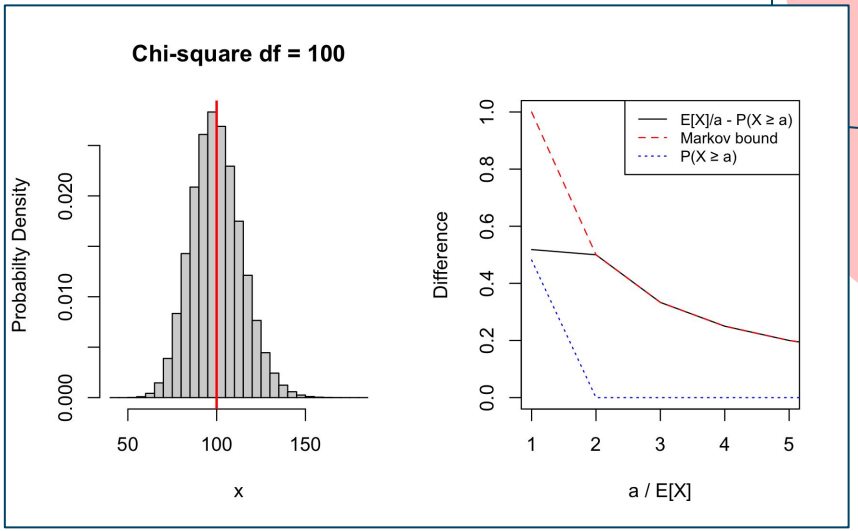


Chi-square Distribution

df = 1

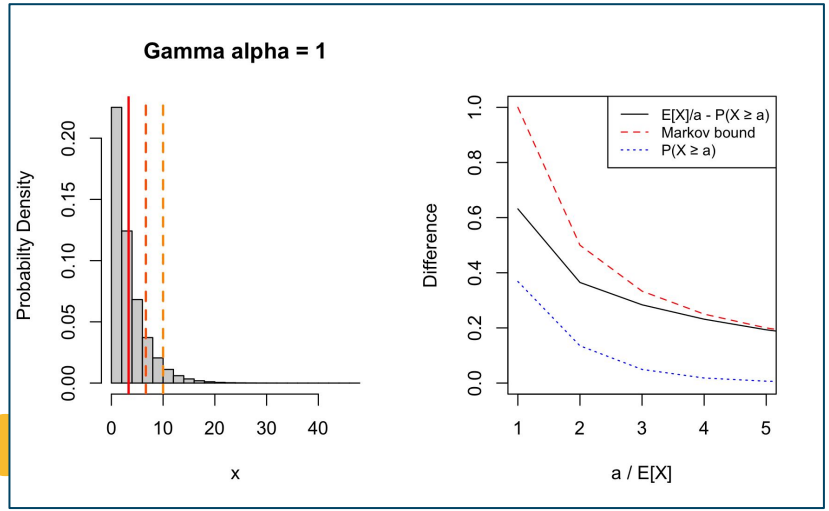


df = 100

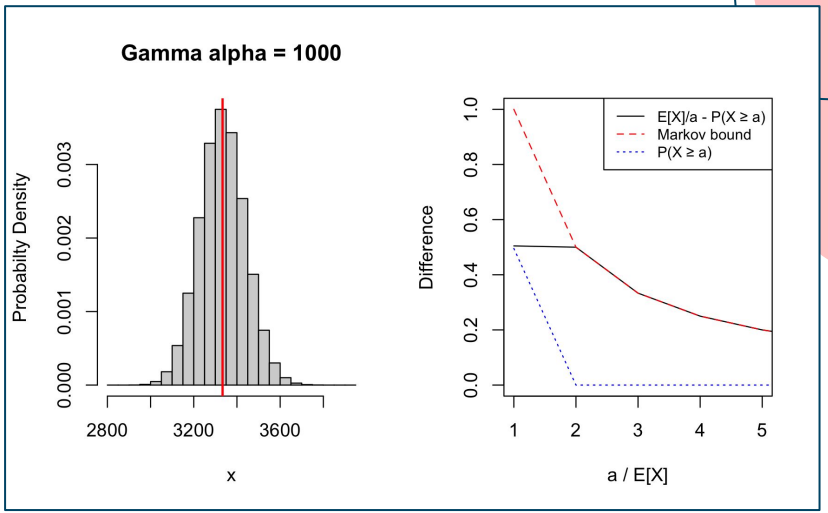


Gamma Distribution

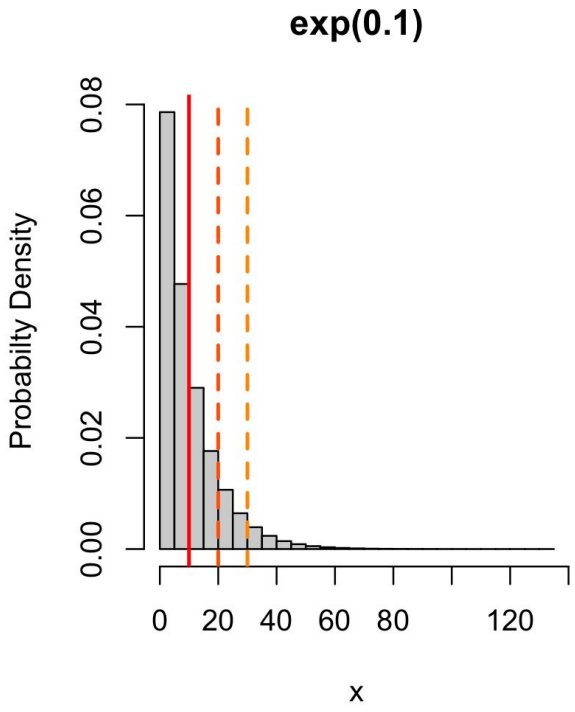
α = 1

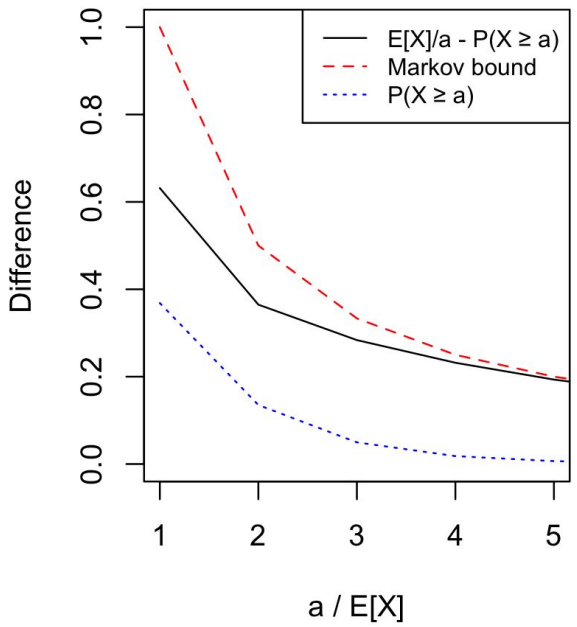


α = 1000

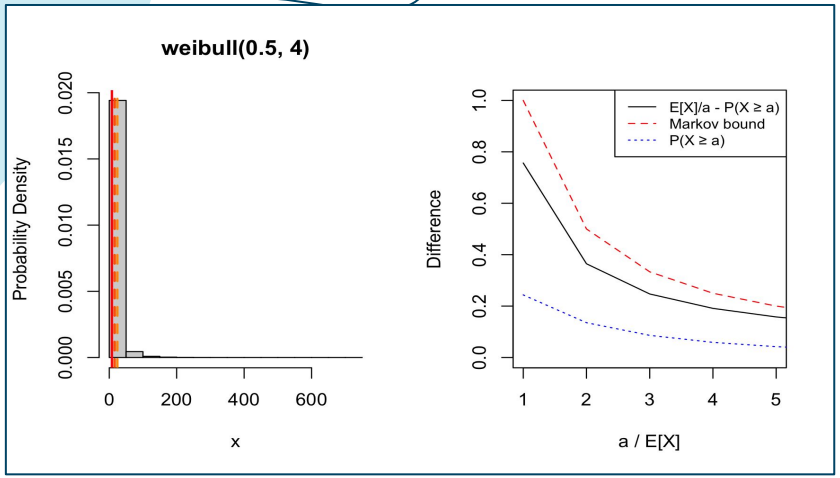


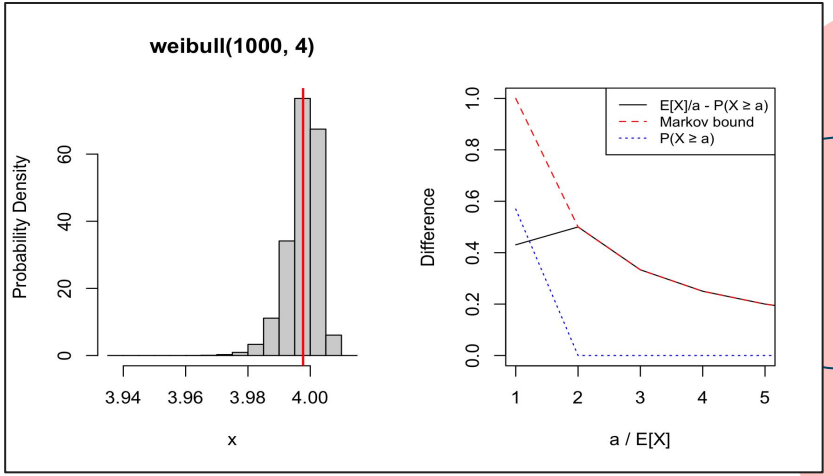
Exponential Distribution

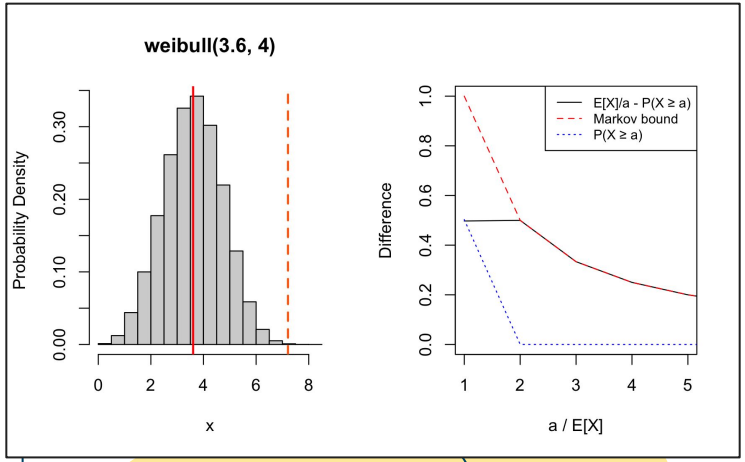




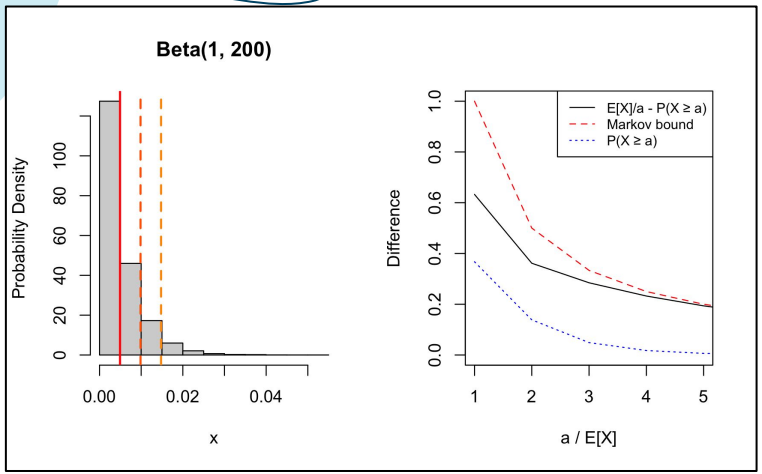
Weibull Distribution

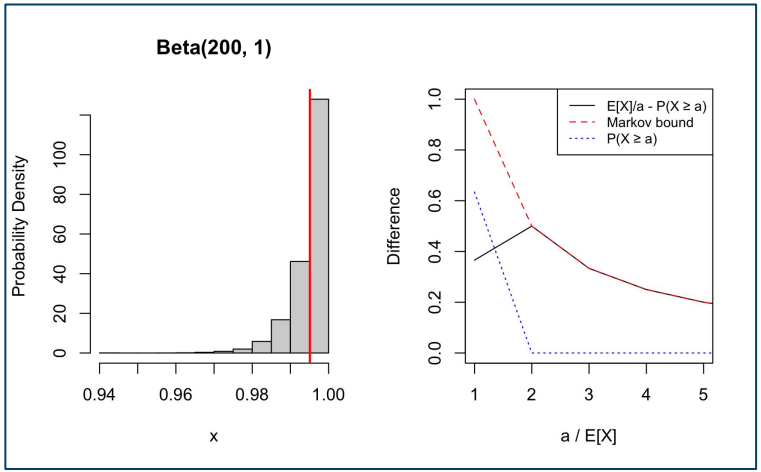


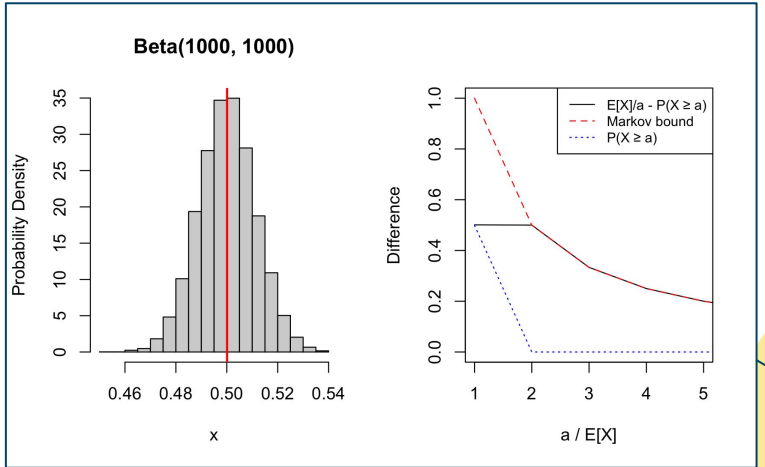


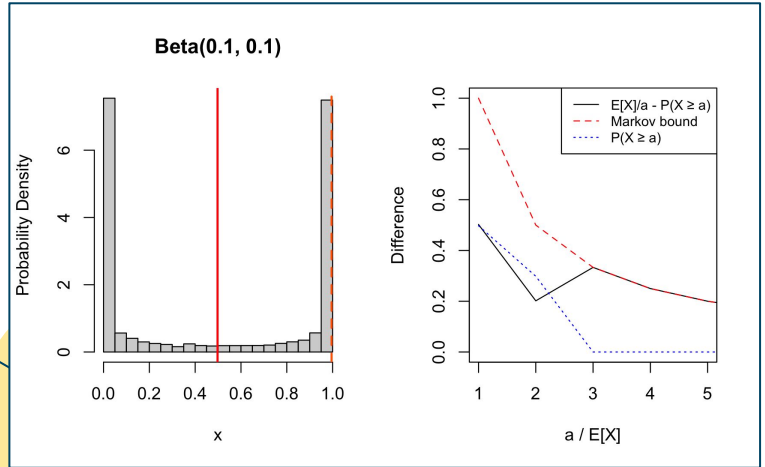


Beta Distribution

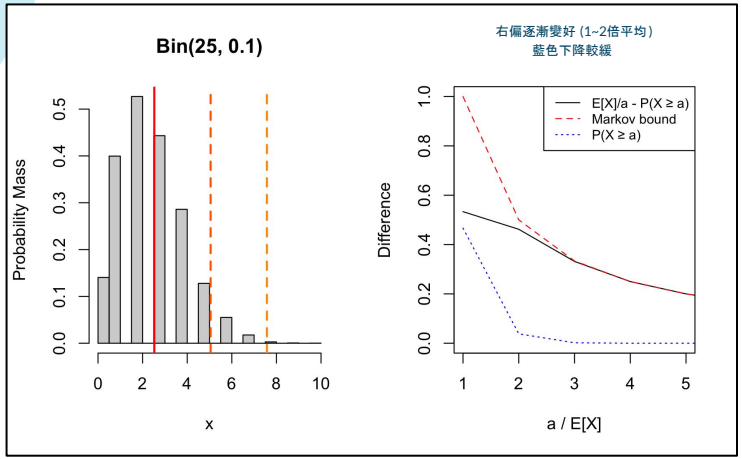


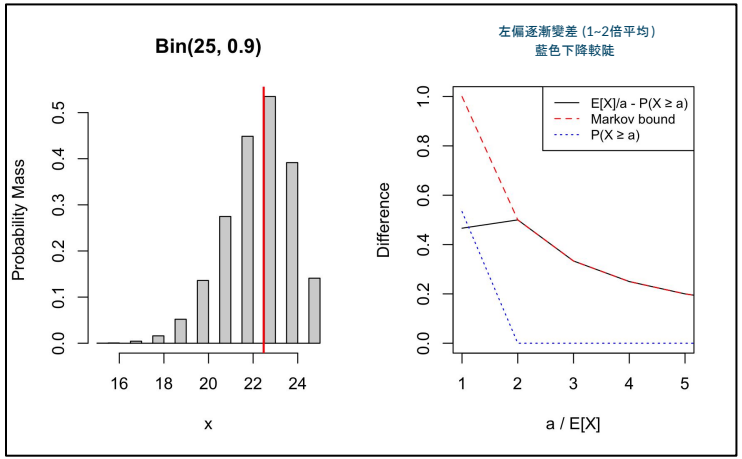


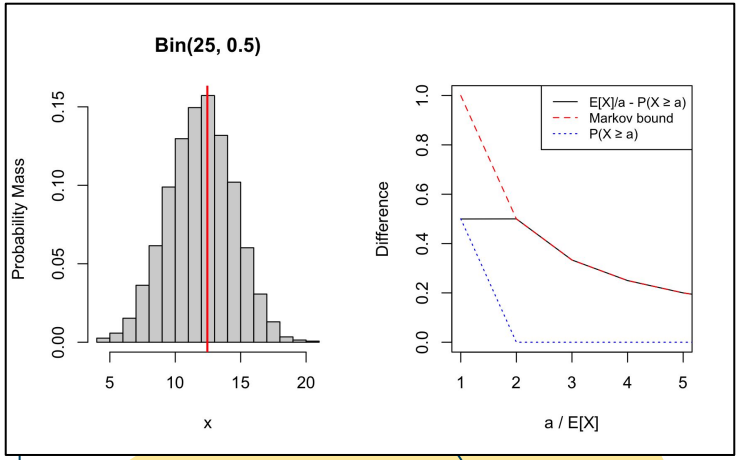




Binomial Distribution

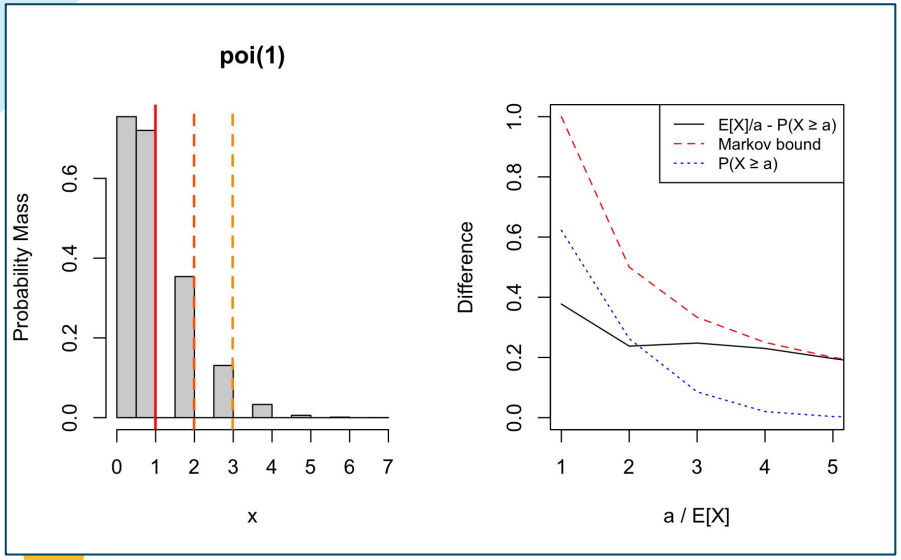


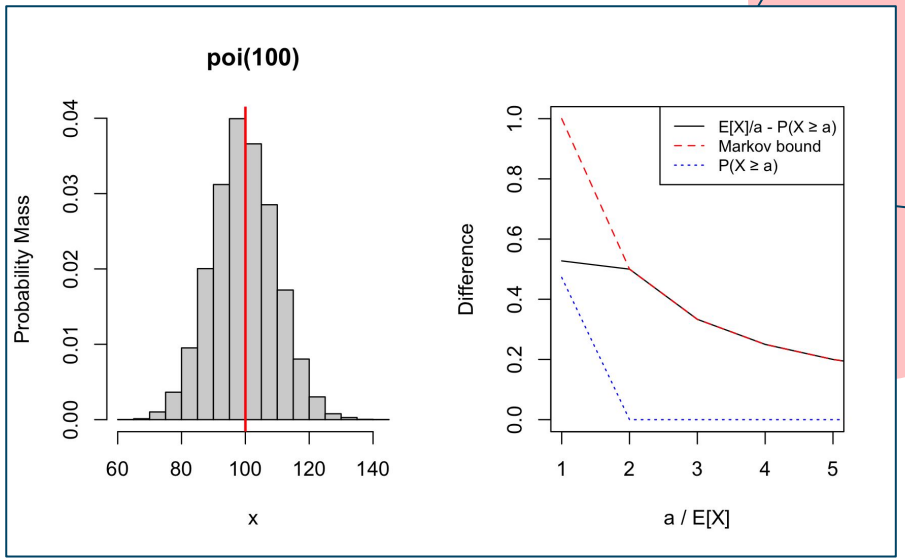




ai = i * mean(X)
ai/E[X] = i
(i 為幾倍的平均數)

Poisson Distribution





ai = i * mean(X)
ai/E[X] = i
(i 為幾倍的平均數)

04

總結

	嚴謹程度 (1~2倍平均數間)	嚴謹程度 (大於2倍平均數)
左偏	逐漸變差	穩定變好
鐘型分佈	保持不變	
右偏	逐漸變好	